Thermal Physics (PH2104) Mid-sem Exam (September 12, 2025)

Answer ALL questions! Marks are indicated in brackets. Total marks =20

1. Obtain the pressure-Temperature (p-T) relationship for *one* mole of an ideal gas undergoing an adiabatic process starting from the first law of thermodynamics. (3)

2. An ideal gas goes through a cycle consisting of an isobaric, adiabatic and isothermal process. The isotherm corresponds to the lowest temperature during the cyclic process. If the maximum temperature (T_2) is ***n*** times the minimum temperature (T_1), draw the p-V diagram and obtain an expression for the efficiency of the cycle. (1+3)

3. Starting from the first law of thermodynamics and using dQ=TdS, derive an expression for the difference in entropy (ΔS) of *one* mole of an ideal gas undergoing an expansion from volume V_A to volume V_B. The temperatures of the states A & B can be taken to be T_A & T_B. For an isothermal expansion process, calculate the difference in entropy if the final volume of the gas is twice the initial volume. (2+1)

4. An ideal gas operates on a cycle, when represented in a p-V diagram is a rectangle. If p_1 and p_2 are the lower and higher pressures respectively and V_1 and V_2 are the corresponding volumes,

(i) Calculate the work done in one complete cycle. (2)

(ii) Indicate in the p-V diagram, in which part heat is absorbed by the gas and in which part heat is released by the gas. Give a brief justification for your answer. Calculate the amount of heat energy (a) absorbed (b) released by the gas and in one cycle. (2+2+2)

(iii) Show that the efficiency of the engine is $\eta = \dfrac{\gamma - 1}{\dfrac{\gamma p_2}{p_2 - p_1} + \dfrac{V_1}{V_2 - V_1}}$ (2)